Fouriers theorem: $x = f(t)$

Statement: " Any single valued, continuous function, what ever complex it may bee can always be represented as the sum of simple harmonic terms; having frequencies which are integral multiple of the frequency of the given periodic function."

Explanation: Let $x = f(t)$ a periodic function having a frequency n; angular frequency $\omega = 2\pi n$.
Fouriers theorem states that:

$$x = f(t) = A_0 \sin(0.\omega t + \psi_0) + A_1 \sin(1.\omega t + \psi_1) + A_2 \sin(2\omega t + \psi_2) + A_3 \sin(3\omega t + \psi_3) + \cdots \quad (1)$$

Equation (1) is the mathematical representation of the Fourier's theorem:
Let us now simplify equation (1):

$$x = A_0 \sin\psi_0 + A_1 \sin\omega t.\cos\psi_1 + A_1 \cos\omega t.\sin\psi_1 + A_2 \sin 2\omega t.\cos\psi_2 + A_2 \cos 2\omega t.\sin\psi_2 + A_3 \sin 3\omega t.\cos\psi_3 + A_3 \cos 3\omega t.\sin\psi_3 + \cdots$$

Put $A_0 \sin\psi_0 = a_0$; $A_1 \sin\psi_1 = a_1$; $A_2 \sin\psi_2 = a_2$
$A_3 \sin\psi_3 = a_3 \cdots$
& $A_1 \cos\psi_1 = b_1$, $A_2 \cos\psi_2 = b_2$; $A_3 \cos\psi_3 = b_3 \cdots$

$$\therefore x = a_0 + b_1 \sin\omega t + a_1 \cos\omega t + b_2 \sin 2\omega t + a_2 \cos 2\omega t + b_3 \sin 3\omega t + a_3 \cos 3\omega t + \cdots$$

or $$\boxed{x = a_0 + \sum_{r=1} a_r \cos r\omega t + \sum_{r=1} b_r \sin r\omega t} \quad (2)$$

Equation (2) is modified representation of Fouriers theorem where a_0, a_r's & b_r's are unknown constants.

<u>We now evaluate the constants</u>:

(i) To find a_0:

Multiplying both sides of equation (2) by 'dt' & integrating between the limits 0 to T

$$\int_0^T x\,dt = \int_0^T a_0\,dt + \int_0^T \sum_{r=1} a_r \cos r\omega t\,dt + \int_0^T \sum b_r \sin r\omega t\,dt$$

or
$$\int_0^T x\,dt = a_0 T + \sum_{r=1}\left[a_r \int_0^T \cos r\omega t\,dt\right] + \sum_{r=1}\left[b_r \int_0^T \sin r\omega t\,dt\right] \quad \text{——— (3)}$$

$$\int_0^T \cos r\omega t\,dt = \int_0^T \sin r\omega t = 0 \quad \text{——— (4)}$$

$$\int_0^T \cos r\omega t\,dt = \left[\frac{\sin r\omega t}{r\omega}\right]_0^T = \frac{1}{r\omega}\left[\sin r\omega T - \sin r\omega \cdot 0\right]$$
$$= \frac{1}{r\omega}\left[\sin 2\pi r - \sin 0\right] = \frac{1}{\omega r}[0-0] = 0$$

$$\int_0^T \sin r\omega t\,dt = \frac{1}{r\omega}\left[\cos r\omega t\right]_T^0 = \frac{1}{r\omega}\left[\cos 0 - \cos 2\pi r\right]$$
$$= \frac{1}{r\omega}[1-1] = 0$$

Putting equation (4) in (3) :-

$$\int_0^T x\,dt = a_0 T + \sum_r a_r \cdot 0 + \sum_r b_r \cdot 0$$

or
$$\boxed{a_0 = \frac{1}{T}\int_0^T x\,dt} \quad \text{——— (5)}$$

(ii) To find a_r :-

Multiplying both sides of the equation by $\cos m\omega t\, dt$ & integrating between the limits $t=0$ to T.

$$\int_0^T x \cos m\omega t\, dt = \int_0^T a_0 \cos m\omega t\, dt + \sum_{r=1} a_r \int_0^T \cos r\omega t \cdot \cos m\omega t\, dt$$

$$+ \sum_{r=1} b_r \int_0^T \sin r\omega t \cdot \cos m\omega t\, dt \quad \text{—(6)}$$

$I_1 = \int_0^T \cos m\omega t\, dt = 0$; where 'm' is an integer (equation $\longrightarrow$ (7)

$$I_2 = \int_0^T \sin r\omega t \cdot \cos m\omega t\, dt = 0 \text{ ——— (8)}$$

$$I_2 = \frac{1}{2} \int_0^T 2\sin r\omega t \cdot \cos m\omega t\, dt = \frac{1}{2} \int_0^T [\sin (r+m)\omega t + \sin(r-m)\omega t]\, dt$$

put $r+m = p$ = integer ; $r-m = q$ = integer

$$I_2 = \frac{1}{2}\left[\int_0^T \sin p\omega t\, dt + \int_0^T \sin q\omega t\, dt\right]$$

$$I_2 = \frac{1}{2}[0+0] = 0$$

$$I_3 = \int_0^T \cos r\omega t \cdot \cos m\omega t\, dt = 0 \quad \text{if } r \neq m$$
$$= \frac{T}{2} \quad \text{if } r = m \quad \Big|\ (9)$$

$r \neq m$ $\quad I_3 = \frac{1}{2} \int_0^T [\cos (r-m)\omega t + \cos (r+m)\omega t]\, dt$

$$I_3 = \frac{1}{2}\left[\int_0^T \cos q\omega t\, dt + \int_0^T \cos p\omega t\, dt\right]$$

or $I_3 = \frac{1}{2}[0+0] = 0$

$r = m$: $I_3 = \frac{1}{2}\int_0^T [1 + \cos p\omega t]\, dt$

$$I_3 = \frac{1}{2}\left[\int_0^T dt + \int_0^T \cos 2m\omega t\, dt\right]$$

dt

$$= \frac{1}{2}[T + 0] = \frac{T}{2}$$

Putting equation (7), (8) & (9) in (6) :-

$$\int_0^T x \cos m\omega t\, dt = a_0 \times 0 + \left[a_1 \times 0 + a_2 \times 0 + \cdots + a_m \cdot \frac{T}{2} + 0 + 0 + \cdots\right] + \sum b_r \cdot 0$$

Or, $a_m = \frac{2}{T}\int_0^T x \cos m\omega t\, dt$

$$\boxed{a_r = \frac{2}{T}\int_0^T x \cos r\omega t\, dt} \quad \text{——— (10)}$$

(iii) To find b_r : Multiplying both sides of equation (2) by '$\sin m\omega t\, dt$' & integrating between the limits 0 to T.

$$\int_0^T x \sin m\omega t\, dt = \int_0^T a_0 \sin m\omega t\, dt + \sum_{r=1} a_r \int_0^T \cos r\omega t . \sin m\omega t\, dt$$

$$+ \sum_{r=1} b_r \int_0^T \sin r\omega t . \sin m\omega t\, dt \quad \text{——— (11)}$$

$$I_1' = \int_0^T \sin m\omega t\, dt = 0 \quad \text{——— (12)} \quad [\text{As in eq}^n\ 4]$$

$$I_2' = \int_0^T \cos r\omega t . \sin m\omega t\, dt = 0 \quad \text{— (13)} \quad [\text{from } 8]$$

$$I_3' = \int_0^T \sin r\omega t . \sin m\omega t\, dt = \begin{cases} 0 & \text{if } r \neq m \\ \frac{T}{2} & \text{if } r = m \end{cases} \quad \text{—(14)}$$

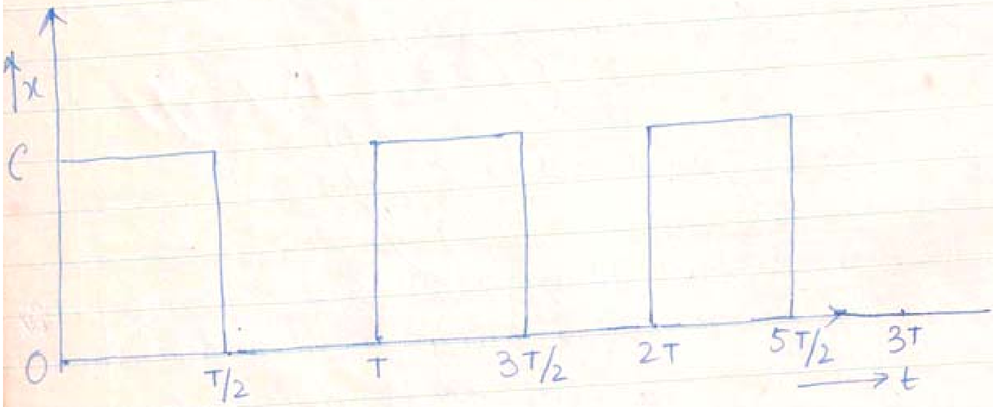
Putting equations (12), (13) and (14) in (11) :-

$$\int_0^T x \sin m\omega t \, dt = a_0 \times 0 + \sum a_r \times 0 + \left[b_1 \times 0 + b_2 \times 0 + \cdots \right.$$
$$\left. \cdots + b_m \frac{T}{2} + b_{m+1} \times 0 + \cdots \right]$$

$$\boxed{b_m = \frac{2}{T} \int_0^T x \sin r\omega t \, dt} \quad \text{—— (15)}$$

Thus the unknown constants can be evaluated.

Application: Applying Fourier's theorem; solve square wave.



$$\left.\begin{array}{ll} x = c & \text{for } 0 \le t \le T/2 \\ x = 0 & \text{for } \frac{T}{2} \le t \le T \end{array}\right| \text{—— (1)}$$

Applying Fouriers theorem; 'x' can be written as

$$x = a_0 + \sum_{r=1} a_r \cos r\omega t + \sum_{r=1} b_r \sin r\omega t \quad \text{—— (2)}$$

where $a_0 = \frac{1}{T} \int_0^T x \, dt$ —— (3)

$a_r = \frac{2}{T} \int_0^T x \cos r\omega t \, dt$ —— (4)

—— (5)

To find a_0 :

$$a_0 = \frac{1}{T}\left[\int_0^{T/2} x\,dt + \int_{T/2}^{T} x\,dt\right]$$

Putting equation (1) :

$$a_0 = \frac{1}{T}\left[\int_0^{T/2} c\,dt + \int_{T/2}^{T} 0.dt\right]$$

or, $$\boxed{a_0 = \frac{c.\pi}{\pi.2} = \frac{c}{2}} \quad \text{—— (6)}$$

To find $a_r = \frac{2}{T}\left[\int_0^{T/2} c\cos r\omega t\,dt + \int_{T/2}^{T} 0.\cos r\omega t\,dt\right]$

or, $$a_r = \frac{2c}{T}\left[\frac{\sin r\omega t}{r\omega}\right]_0^{T/2} = \frac{2c}{r\omega T}\left[\sin r\omega.\frac{T}{2} - \sin r\omega .0\right]$$

$$a_r = \frac{2c}{r.2\pi}\left[\sin \pi r - 0\right] = \frac{c}{\pi r}\left[0-0\right] = 0 \quad \text{— (7)}$$

To find b_r : $\frac{2}{T}\left[\int_0^{T/2} c\sin r\omega t\,dt + \int_{T/2}^{T} 0.\sin r\omega t\,dt\right]$

or, $$b_r = \frac{2c}{T}\left[-\frac{\cos r\omega t}{r\omega}\right]_0^{T/2} = \frac{2c}{r\omega T}\left[\cos 0 - \cos r\omega.\frac{T}{2}\right]$$

or, $$b_r = \frac{2c}{r.2\pi}\left[1 - \cos \pi r\right]$$

If $r = 2, 4, 6, 8 \ldots\ldots$ $\cos \pi r = +1$ $b_r = 0$

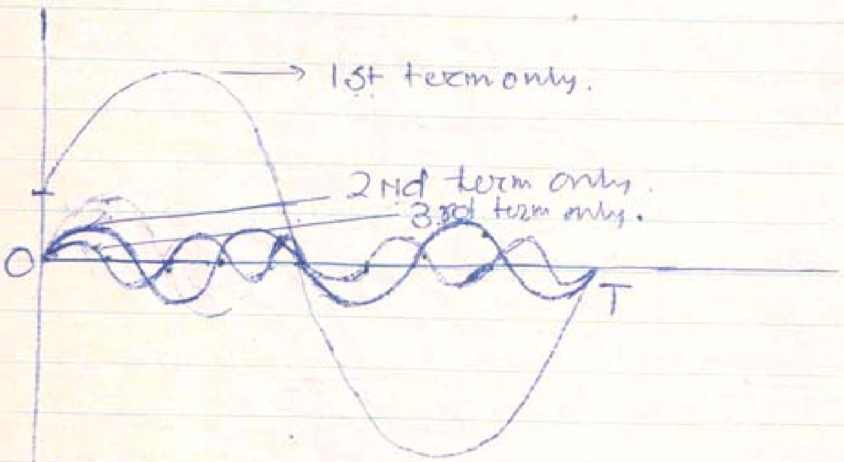
If $r = 1, 3, 5, 7 \ldots\ldots$ $\cos \pi r = (-1)$ $\boxed{b_r = \frac{2c}{\pi r}}$

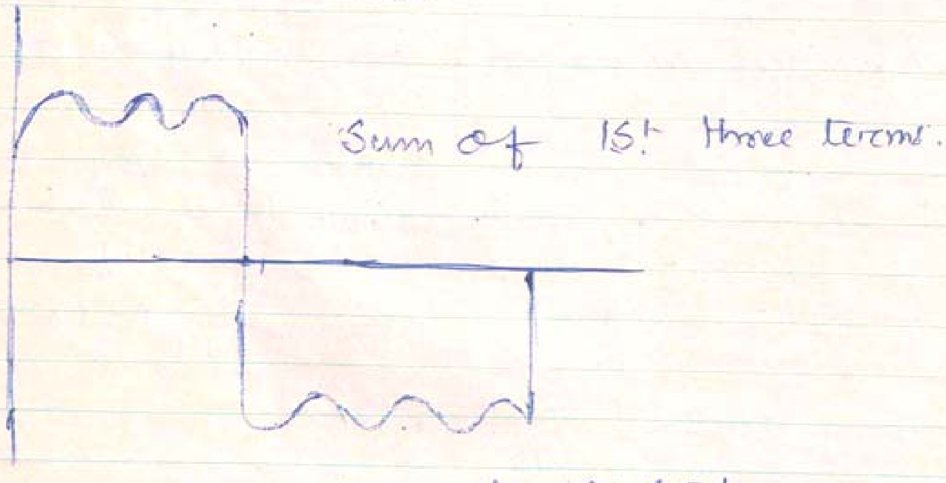
$b_r = 0$ | — r is even | — (8)

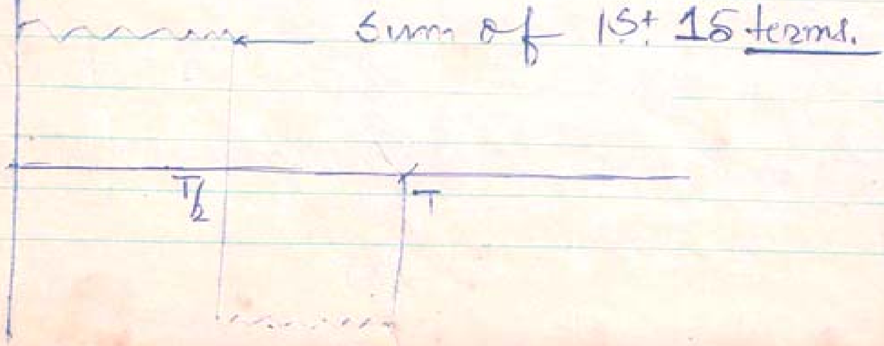
Putting equations ⑥, ⑦ & ⑧ in ②:-

$$x = \frac{c}{2} + \sum_{r=1} 0 \cdot \cos r\omega t + \sum \frac{2c}{\pi r} \sin r\omega t \quad r = odd$$

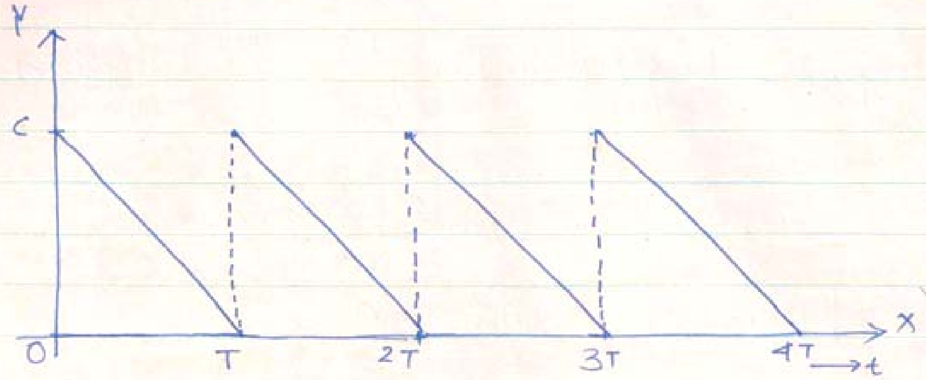
$$\text{or, } x = \frac{c}{2} + \frac{2c}{\pi}\left[\frac{1}{1}\sin\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{5}\sin 5\omega t + \frac{1}{7}\sin 7\omega t + \cdots\right] \quad (9)$$







Saw tooth wave



$x = f(t) = c\left(1 - \frac{t}{T}\right)$ - The given periodic function using Fourier's theorem; this complex periodic function can be represented as

$$x = a_0 + \sum_{r=1}^{\infty} a_r \cos r\omega t + \sum_{r=1}^{\infty} b_r \sin r\omega t \quad \text{—— (1)}$$

Where $a_0 = \frac{1}{T}\int_0^T x\,dt$ —— (2)

$$a_r = \frac{2}{T}\int_0^T x \cos r\omega t\,dt \quad \text{—— (3)}$$

$$b_r = \frac{2}{T}\int_0^T x \sin r\omega t\,dt \quad \text{—— (4)}$$

Putting the value of x in equation (2)

$$a_0 = \frac{1}{T}\int_0^T c\left(1 - \frac{t}{T}\right)dt = \frac{c}{T}\left[\int_0^T dt - \frac{1}{T}\int_0^T t\,dt\right]$$

$$a_0 = \frac{c}{T}\left[T - \frac{1}{\cancel{T}}\cdot\frac{T^2}{2}\right] \text{ or, } a_0 = \frac{c}{\cancel{T}}\cdot\frac{\cancel{T}}{2} = \frac{c}{2} \quad \text{—(5)}$$

Putting the value of x in equation (3):

$$a_r = \frac{2c}{T}\int_0^T \left(1 - \frac{t}{T}\right)\cos r\omega t\,dt$$

$$\text{or, } a_r = \frac{2c}{T}\left[\int_0^T \cos r\omega t\,dt - \frac{1}{T}\int_0^T t\cos r\omega t\,dt\right]$$

Integrating the second term by parts:

$$a_r = \frac{2c}{T}\left[0 - \frac{1}{T}\left\{\left(t\,\frac{\sin r\omega t}{r\omega}\right)_0^T - \int_0^T \frac{\sin r\omega t}{r\omega}\,dt\right.\right]$$

$$\text{or, } a_r = \frac{2c}{T^2}\left[-\frac{1}{r\omega}\left\{T\sin r\omega T - 0\cdot\sin r\omega\cdot 0\right\} - \frac{1}{r\omega}\cdot 0\right]$$

$\boxed{\omega T = 2\pi}$ $\sin r\omega T = \sin 2\pi r = 0$

$$\therefore a_r = \frac{2c}{T^2}\left[0 - \frac{1}{\omega r}\cdot 0 + \frac{1}{\omega r}\cdot 0\right] = 0 \quad \text{—— (6)}$$

Putting the value of 'x' in equation (4) :

$$b_r = \frac{2c}{T}\int_0^T \left(1 - \frac{t}{T}\right)\sin r\omega t\,dt$$

$$= \frac{2c}{T}\left[\underbrace{\int_0^T \sin r\omega t\,dt}_{=0} - \frac{1}{T}\int_0^T t\sin r\omega t\,dt\right]$$

Integrating the 2nd term by parts :-

$$b_r = \frac{2c}{T}\left[0 - \frac{1}{T}\left\{t\left(-\frac{\cos r\omega t}{r\omega}\right) + \frac{1}{r\omega}\int_0^T \cos r\omega t\,dt\right]_0^T\right.$$

$$\text{or, } b_r = \frac{2c}{T^2}\left[\frac{1}{r\omega}\left\{T\cos r\omega T - 0\cdot\cos r\omega\cdot 0\right\}\right]$$

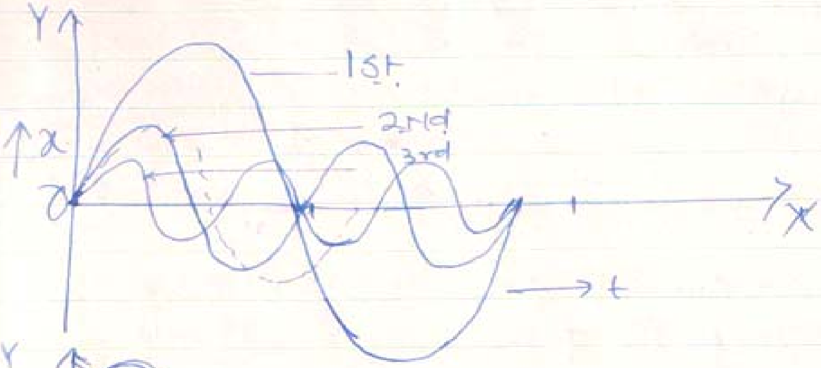
$$\text{or, } b_r = \frac{2c}{r\omega T\cdot T}\left[T\cos 2\pi r\right] \qquad [\because \omega T = 2\pi]$$

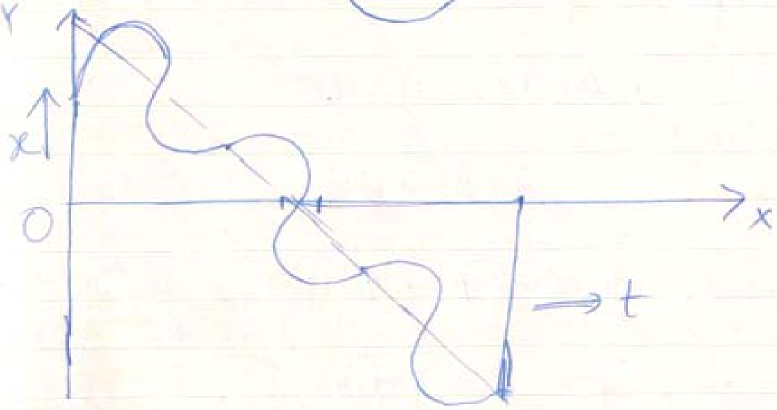
$$\text{or, } b_r = \frac{2c}{r\cdot 2\pi}\times\frac{1}{T}\cdot T \quad \text{or } \boxed{b_r = \frac{c}{\pi r}} \text{ —— (7) } r = 1,2,3\ldots$$

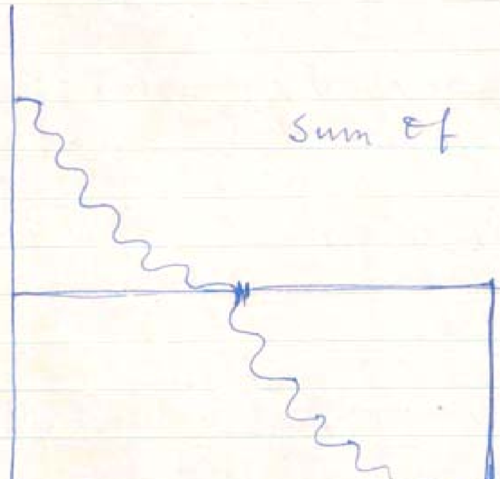
Putting equations (5), (6) & (7) in (1) :-

$$x = \frac{c}{2} + \sum_{r=1}^{\infty}\frac{c}{\pi r}\sin r\omega t$$

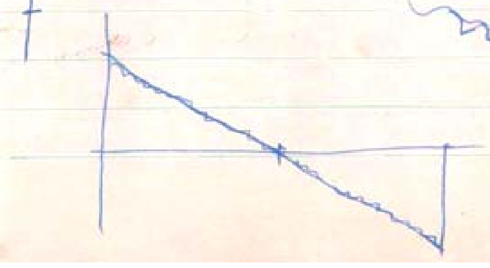
$$x = \frac{c}{2} + \frac{c}{\pi}\left[\sin\omega t + \frac{1}{2}\sin 2\omega t + \frac{1}{3}\sin 3\omega t + \cdots\right]$$







Sum of 1st 15 terms.



Sum of 1000 terms.

Rough :

$$\sin 30^\circ = \sin(2\pi + 30^\circ)$$

$$f(x) = f(x+2l) \qquad 2l = 2\pi.$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\phi x + \sum_{n=0}^{\infty} b_n \sin\phi x.$$

We chose the value of 'ϕ' in such a way that

$$f(x) = f(x+2l)$$

$$f(x+2l) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\phi(x+2l) + \sum_{n=1}^{\infty} b_n \sin\phi(x+2l)$$

$-l$ $\qquad$ $+l$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(\phi x + \phi 2l) + \sum_{n=1}^{\infty} b_n \sin(\phi x + \phi 2l)$$

If $\phi \cdot 2l = 2n\pi$ $\quad \therefore$ $\boxed{\phi = \frac{n\pi}{l}}$

$$\cos(\phi x + \phi \cdot 2l) = \cos(2n\pi + \phi x) = \cos\phi x$$

$$\sin(\phi x + \phi \cdot 2l) = \sin(2n\pi + \phi x) = \sin\phi x$$

$$\therefore f(x+2l) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\phi x + \sum_{n=1}^{\infty} b_n \sin\phi x.$$

$$= f(x)$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi}{l} x + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi}{l} x,$$

Fourier's Theorem and Fourier Integral:

This theorem states that: "Any single valued function $f(x)$ defined on the interval $[-\pi, \pi]$ may be represented over the interval by the trigonometric series."

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \quad \text{——(1)}$$

The expansion coefficients are given by Euler's formula.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \cos nx \, dx \quad \text{——(2)} \quad n = 0, 1, 2, 3 ..$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{+\pi} f(x) \sin nx \, dx \quad \text{——(3)} \quad n = 1, 2, 3 ..$$

If we put, the values of coefficients from (2) and (3) in (1) the representation of $f(x)$ is known as Fourier Series.

We now consider the change in interval from $[-\pi, \pi]$ to $[-l, l]$

let $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \phi x + \sum_{n=1}^{\infty} b_n \sin \phi x \quad \text{—(4)}$$

Where 'ϕ' is so chosen that $f(x) = f(x+2l)$

i.e the function $f(x)$ becomes periodic over an interval '$2l$'.

replacing x by $(x+2l)$ in (4):-

$$f(x+2l) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\phi(x+2l) + \sum_{n=1}^{\infty} b_n \sin\phi(x+2l)$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(\phi 2l + \phi x) + \sum_{n=1}^{\infty} b_n \sin(\phi \cdot 2l +$$

If we put $\boxed{\phi \cdot 2l = 2n\pi}$, then above equation becomes:

$$f(x+2l) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(2n\pi + \phi x) + \sum_{n=1}^{\infty} b_n \sin(2n\pi + \phi x$$

$$\text{or } f(x+2l) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\phi x + \sum_{n=1}^{\infty} b_n \sin\phi x.$$

$$\text{or } \boxed{f(x+2l) = f(x)}$$

Hence the function $f(x)$ becomes periodic over the interval $[-l, l]$ only if $\boxed{\phi = \frac{n\pi}{l}}$

Hence equation (4) can be written as:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l} + \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{l} \quad \text{— (5)}$$

Following equation (2) and (3) the coefficient becomes :-

$$a_n = \frac{1}{l}\int_{-l}^{+l} f(x) \cos\left(\frac{n\pi x}{l}\right) dx \quad \text{— (6)} \quad n = 0, 1, 2, 3 \dots$$

$$b_n = \frac{1}{l}\int_{-l}^{+l} f(x) \sin\left(\frac{n\pi x}{l}\right) dx \quad \text{— (7)}$$

We now define Fourier Integral:

$$f(x) = \frac{1}{2} \cdot \frac{1}{l} \int_{-l}^{+l} f(t)\,dt + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi t}{l}$$

$$+ \sum_{n=1}^{\infty} b_n \sin\frac{n\pi t}{l}.$$

$$\sum a_r \cos r\omega t = \sum a_n \cos n \frac{2\pi}{2l} \times t = \sum a_n \cos\frac{n\pi t}{l}$$

$T \longrightarrow 2l$ because the interval $(-l, l)$
$2l$ is periodicity.

So that the expansion involves all values of 'x' in the interval $[-l \text{ to } +l]$

$$\therefore f(x) = \frac{1}{2l}\int_{-l}^{+l} f(t)\,dt + \sum_{n=1}^{\infty}\left(\frac{1}{l}\int_{-l}^{+l} f(t)\cos\frac{n\pi t}{l}\,dt\right)\cos\frac{n\pi x}{l}$$

$$+ \sum_{n=1}^{\infty}\left(\frac{1}{l}\int_{-l}^{+l} f(t)\sin\frac{n\pi t}{l}\,dt\right)\cdot\sin\frac{n\pi x}{l}$$

$$\text{or, } f(x) = \frac{1}{2l}\int_{-l}^{+l} f(t)\,dt + \frac{1}{l}\sum_{n=1}^{\infty}\int_{-l}^{+l} f(t)\cos\left\{\frac{n\pi}{l}(t-x)\right\}dt$$

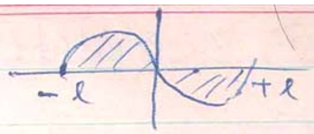
$$\boxed{\because \cos A \cdot \cos B + \sin A \cdot \sin B = \cos(A-B)}$$

$$f(x) = \frac{1}{2l}\int_{-l}^{+l} f(t)\,dt + \frac{1}{l}\sum_{n=1}^{\infty}\int_{-l}^{+l} f(t)\cos\frac{n\pi}{l}(t-x)\,dt$$

If we set $\frac{n\pi}{l} = u$ —ⓐ where $\Delta u = \frac{\pi}{l}$ or $\frac{1}{l} = \frac{\Delta u}{\pi}$ ⟶ ⓑ

where $\boxed{\Delta n = 1}$

lso $\frac{1}{2l}\int_{-l}^{+l} f(t)\,dt = 0$ —— (11)



Because when the sinusoidal function $f(t)$ integrated over the limit $-l$ to $+l$; the sum total of the two half cycles being equal and opposite is zero].

Putting equations (9), (10) and (11) in (8) :-

$$f(x) = 0 + \sum_{u=\pi/l}^{\infty} \frac{\Delta u}{\pi} f(t)\cos[u(t-x)]\,dt$$

Or $$f(x) = \frac{1}{\pi}\sum_{u=\pi/l}^{\infty} \Delta u \int_{-\infty}^{+\infty} f(t)\cos[u(t-x)]\,dt$$

Put $$\sum_{u=\pi/l}^{\infty} \Delta u = \int_{0}^{\infty} du$$

$$\therefore f(x) = \frac{1}{\pi}\int_{0}^{\infty} du \int_{-\infty}^{+\infty} f(t)\cos[u(t-x)]\,dt \quad \text{—— (12)}$$

Equation (12) is known as Fourier's Integral.